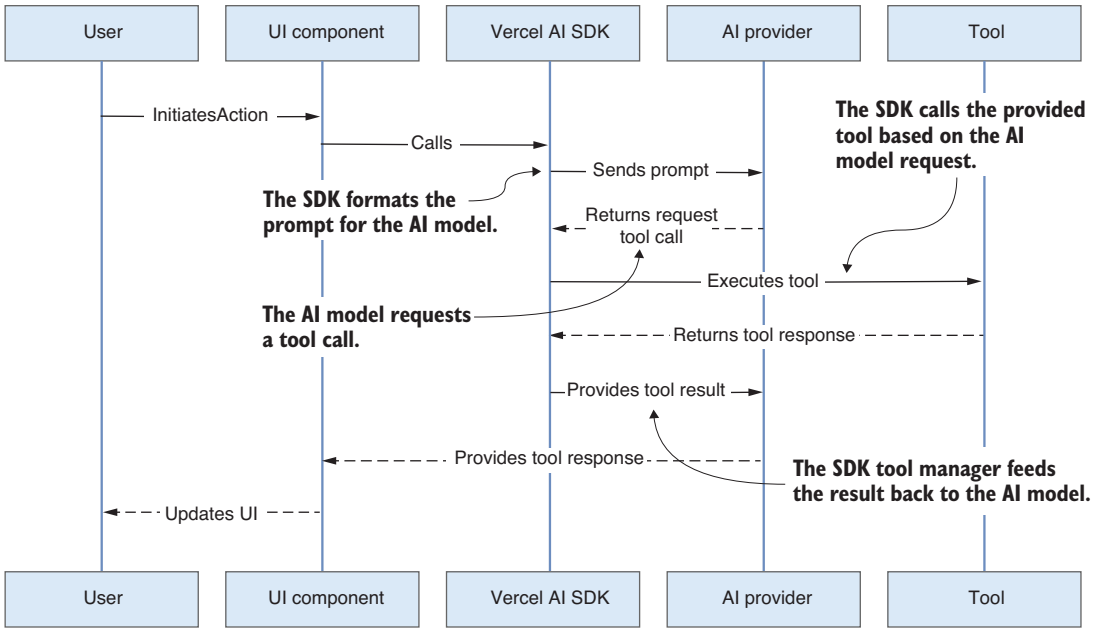


Figure 4.14 The process of tool calling in an AI-powered application using the Vercel AI SDK. This shows how a user action initiates a prompt, which is formatted by the SDK and sent to the AI provider. The AI model may request a tool call, which is then executed by the SDK. The tool's response is fed back to the AI model through the SDK, and finally, the result is used to update the UI for the user.

tool manager, or Vercel AI SDK in our case, comes into play. It interprets the AI's request, executes the appropriate custom tool (which might involve API calls, database queries, or local functions), and collects the result. The SDK then sends this tool result back to the external AI model service. This workflow allows the AI model to use external tools and data while keeping the execution of these tools under our application's control, maintaining a separation between the AI processing and our specific toolset.

Before we continue, I should mention a simpler way to do this. We could also put the response of the external API call directly into the prompt instead of relying on the tool manager. For example, say we make an API call to get the weather data and extract the relevant information from the API response. We can directly inject the response to the prompt and ask the model to provide us with the answer. The following listing shows an example.

Listing 4.13 Getting weather info and displaying it

```
  const {city, temperature, weatherDescription,
humidity} = await getWeatherInfo(city);
```

Calls the weather info API to get the weather report